Olympiad Test Paper — Science (Class 7)

Chapter 5: Acids, Bases and Salts

Subject	Science (NCERT Class 7)
Chapter	5 — Acids, Bases and Salts
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Remember the litmus rule: blue litmus turns red in acid, red litmus turns blue in a base, and neither changes in a neutral solution. Turmeric and china-rose are natural indicators; phenolphthalein is a synthetic one. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

---

1. Which property is shared by **all acids**?

(A) they taste sour
(B) they feel soapy
(C) they turn red litmus blue
(D) they taste bitter

2. A liquid feels soapy to touch and tastes bitter. It is most likely:

(A) an acid
(B) a base
(C) a neutral solution
(D) common salt water

3. Which acid is present in curd (and gives sour milk its sourness)?

(A) acetic acid
(B) citric acid
(C) lactic acid
(D) tartaric acid

4. The acid present in vinegar is:

(A) acetic acid
(B) formic acid
(C) tartaric acid
(D) hydrochloric acid

5. Tamarind owes its sour taste to:

(A) lactic acid
(B) citric acid
(C) acetic acid
(D) tartaric acid

6. Which of the following is a **base**?

(A) lemon juice
(B) milk of magnesia
(C) vinegar
(D) curd

7. The acid found in our stomach that helps digest food and kill germs is:

(A) hydrochloric acid (B) sulphuric acid
(C) citric acid (D) carbonic acid

8. Which of these is obtained from a lichen and used as a natural indicator?

(A) phenolphthalein (B) methyl orange
(C) litmus (D) baking soda

9. A student dips **blue** litmus paper into a sample and it turns **red**. The sample is:

(A) basic (B) neutral
(C) acidic (D) a salt solution

10. A student dips **red** litmus paper into a sample and there is **no colour change**. Dipping blue litmus also shows no change. The sample is:

(A) acidic (B) basic
(C) neutral (D) strongly acidic

11. Soap solution is rubbed on a turmeric stain on a white shirt. The turmeric spot turns:

(A) red (B) blue
(C) green (D) colourless

12. Phenolphthalein is added to a colourless solution and it turns **pink**. The solution is:

(A) acidic (B) basic
(C) neutral (D) common salt water

13. The reaction between an acid and a base is called:

(A) neutralisation (B) oxidation
(C) evaporation (D) condensation

14. When an acid completely neutralises a base, the products are:

(A) salt and hydrogen (B) acid and water
(C) base and water (D) salt and water

15. A red ant's sting injects an acid. The best thing to rub on the sting for relief is:

(A) lemon juice (B) vinegar
(C) baking soda paste (D) more honey

16. A farmer's field has soil that is too acidic for crops. To correct it, the farmer should add:

(A) vinegar (B) lemon juice
(C) quicklime or slaked lime (D) more acidic fertiliser

17. A factory's waste water is acidic before being released into a river. To make it safe, the factory treats it with:

(A) more acid | (B) a base such as slaked lime
(C) common salt | (D) sugar

18. When a person has acidity (too much acid in the stomach), a doctor advises an **antacid**, which works because it is:

(A) a strong acid | (B) a sweet sugar
(C) common salt | (D) a mild base that neutralises the excess acid

19. Common table salt (sodium chloride) is formed by neutralising:

(A) HCl with NaOH | (B) citric acid with vinegar
(C) HCl with sulphuric acid | (D) NaOH with milk of magnesia

20. Distilled water turns:

(A) blue litmus red | (B) red litmus blue
(C) neither litmus paper (no change) | (D) both litmus papers green

21. China rose (gudhal) indicator turns **light green** in one type of solution. That solution is:

(A) acidic | (B) neutral
(C) made of common salt | (D) basic

22. Phenolphthalein is added separately to lemon juice and to soap solution. The colours seen are, respectively:

(A) pink and colourless | (B) colourless and pink
(C) pink and pink | (D) colourless and colourless

23. Three samples are tested with **blue** litmus only: lemon juice, soap solution, sugar solution. In how many does the blue litmus turn red?

(A) 0 | (B) 1
(C) 2 | (D) 3

24. Which gas, dissolving in rain water, is the main cause of acid rain?

(A) oxygen | (B) nitrogen
(C) sulphur dioxide | (D) carbon monoxide

25. Sodium hydroxide turns:

(A) red litmus blue | (B) blue litmus red
(C) neither litmus | (D) turmeric paper colourless

26. A liquid does **not** change the colour of either litmus paper but turns phenolphthalein pink. This result is:

(A) consistent — the liquid is acidic | (B) consistent — the liquid is neutral
(C) contradictory — phenolphthalein pink means basic, so litmus should change
(D) impossible to interpret without taste

27. China rose indicator is added to two test tubes. In tube 1 it turns dark pink (magenta); in tube 2 it turns green. Tubes 1 and 2 contain, respectively:

(A) a base and an acid (B) two acids

(C) an acid and a base (D) two bases

28. A spoon of baking soda is added to a glass of vinegar and it fizzes. This shows that:

(A) a base is reacting with an acid (neutralisation)

(B) baking soda is an acid (C) vinegar is a base

(D) nothing chemical is happening

29. Which of the following correctly lists an acid, a base and a neutral substance, in that order?

(A) lemon juice, soap, distilled water (B) soap, lemon juice, salt water

(C) distilled water, lemon juice, soap (D) baking soda, vinegar, lemon juice

30. The acid injected by a stinging bee or ant is:

(A) hydrochloric acid (B) formic acid

(C) tartaric acid (D) carbonic acid

31. Milk of magnesia (magnesium hydroxide) is taken for indigestion. Its taste and feel are:

(A) sour and rough (B) sweet and sticky

(C) salty and oily (D) bitter and soapy

32. A solution turns red litmus blue. Which indicator result would you also expect from the **same** solution?

(A) it turns blue litmus red (B) it turns turmeric paper red

(C) it leaves phenolphthalein colourless (D) it turns turmeric green

33. Which one of these is a **synthetic** (lab-made) indicator, not a natural one?

(A) litmus (B) turmeric

(C) china rose (D) phenolphthalein

34. Orange and other citrus fruits are sour because they contain:

(A) acetic acid (B) citric acid

(C) lactic acid (D) carbonic acid

35. When you mix an acid and a base in just the right amounts, the final mixture's effect on litmus is:

(A) turns blue litmus red (B) turns red litmus blue

(C) no change in either litmus (D) turns both litmus papers green

36. Ammonia is found in many window-cleaning liquids and turns red litmus blue. Ammonia is therefore:

(A) acidic
(B) basic
(C) neutral
(D) a salt

37. A child mixes equal "neutralising" amounts of hydrochloric acid and sodium hydroxide. Apart from water, the product formed is:

(A) sodium chloride
(B) sodium hydroxide
(C) hydrochloric acid
(D) sodium bicarbonate

38. During a neutralisation reaction between an acid and a base, you would also notice that:

(A) heat is released and the mixture warms up
(B) the mixture freezes
(C) the mixture turns into a metal
(D) a strong acid smell increases

39. Four solutions are tested with blue litmus. Lemon juice → red, soap → no change, distilled water → no change, vinegar → red. How many of the four are acidic?

(A) 1
(B) 2
(C) 3
(D) 4

40. Turmeric paper is rubbed and stays **yellow** when touched with a drop of a sample. The sample is:

(A) basic
(B) not basic (acidic or neutral)
(C) definitely an acid
(D) definitely neutral

41. A gardener finds the soil is too **basic (alkaline)** for a plant that likes slightly acidic soil. A sensible treatment is to add:

(A) slaked lime
(B) decaying organic matter (compost), which is mildly acidic
(C) more sodium hydroxide
(D) baking soda

42. Which statement about indicators is **correct**?

(A) phenolphthalein is pink in acids
(B) blue litmus turns red in bases
(C) china rose stays the same colour in acids and bases
(D) turmeric turns red in bases

43. ENO (a fizzy antacid) contains an acid (citric acid) and a base (sodium bicarbonate). When dropped in water it fizzes because:

(A) the acid evaporates
(B) the water boils
(C) the base dissolves silently with no reaction
(D) the acid and base neutralise, releasing a gas

44. A solution turns blue litmus red **and** keeps phenolphthalein colourless. The solution is:

(A) basic
(B) neutral
(C) impossible to have both results
(D) acidic

45. Which of the following is a property of a **base** but **not** of an acid?

(A) it turns blue litmus red (B) it tastes sour

(C) it feels soapy and turns red litmus blue(D) it tastes sour and feels rough

46. A label-less bottle contains a liquid that tastes sour and turns blue litmus red. It is safe to conclude the liquid is:

(A) an acid (B) a base

(C) neutral (D) common salt solution

47. Lemon juice is added drop by drop to a soap solution that has phenolphthalein in it (currently pink). As enough lemon juice is added, the pink colour:

(A) gets darker (B) turns blue

(C) stays exactly the same (D) disappears (turns colourless)

48. Which of these everyday problems is solved by a **neutralisation** reaction?

(A) a torch glowing (B) ice melting in the sun

(C) a fan spinning (D) relieving an ant sting with baking soda

49. The acid used heavily in industry — in fertilisers, car batteries and detergents — is:

(A) citric acid (B) acetic acid

(C) sulphuric acid (D) lactic acid

50. Three liquids are tested. Liquid X turns red litmus blue, liquid Y turns blue litmus red, liquid Z changes neither. Which liquid is the base?

(A) X (B) Y

(C) Z (D) none of them

51. A sample turns turmeric paper **red**. The same sample is most likely to also:

(A) turn blue litmus red (B) turn red litmus blue

(C) leave both litmus papers unchanged (D) taste sour

52. Sugar dissolved in water is tested. The expected results are:

(A) blue litmus → red (B) red litmus → blue

(C) phenolphthalein → pink (D) no change in either litmus (neutral)

53. Which pairing of substance and family is **wrong**?

(A) baking soda — base (B) vinegar — acid

(C) sodium hydroxide — acid (D) lemon juice — acid

54. Acid rain damages marble and limestone monuments such as the Taj Mahal because:

(A) the marble is acidic and reacts with the rain's base

(B) marble (a base, calcium carbonate) is eaten away by the acid in the rain

(C) the rain simply washes away the dust(D) marble reflects sunlight and cracks

55. A solution is neutral. Which set of results matches?

(A) blue litmus red, phenolphthalein pink(B) red litmus blue, turmeric red

(C) no litmus change, phenolphthalein colourless

(D) blue litmus red, turmeric red

56. A pinch of chalk powder (calcium carbonate, a mild base) is added to a beaker of dilute acid. You would expect:

(A) nothing to happen (B) the chalk to make the acid stronger

(C) the mixture to turn blue litmus red even more strongly

(D) the acid to be partly neutralised, with fizzing

57. Methyl orange is added to a basic solution. Knowing it is **yellow in bases and red in acids**, the colour seen is:

(A) red (B) pink

(C) colourless (D) yellow

58. A student claims: "If a liquid does not change litmus, it must be water." The best response is:

(A) correct — only water leaves litmus unchanged

(B) correct — sugar water turns litmus blue(C) wrong — salt water turns blue litmus red

(D) wrong — any neutral solution (e.g. salt water, sugar water) also leaves litmus unchanged

59. A drop of china rose indicator is added to a solution and it turns **dark pink (magenta)**, while the same indicator in another solution turns **green**. Which list correctly names the two solutions?

(A) acid then base (B) base then acid

(C) neutral then acid (D) base then neutral

60. A spoon of baking soda is taken to relieve a sour, acidic stomach. The chemistry of the relief is best described as:

(A) a base neutralising the excess acid to form salt and water

(B) an acid neutralising an acid (C) a base making the stomach more acidic

(D) sugar coating the stomach lining

End of question paper. Maximum marks: 240. Marking: +4 correct, –1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 5: Acids, Bases and Salts

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	D	31	D	41	B	51	B
2	B	12	B	22	B	32	B	42	D	52	D
3	C	13	A	23	B	33	D	43	D	53	C
4	A	14	D	24	C	34	B	44	D	54	B
5	D	15	C	25	A	35	C	45	C	55	C
6	B	16	C	26	C	36	B	46	A	56	D
7	A	17	B	27	C	37	A	47	D	57	D
8	C	18	D	28	A	38	A	48	D	58	D
9	C	19	A	29	A	39	B	49	C	59	A
10	C	20	C	30	B	40	B	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) The defining taste of every acid is sour (citric in lemon, acetic in vinegar, lactic in curd). Soapy feel and bitter taste belong to bases; turning red litmus blue is also a base property — so the only thing common to all acids here is the sour taste.

2. (B) Soapy feel + bitter taste are the two classic signatures of a base. Acids are sour, neutral and salt solutions are tasteless and not soapy.

3. (C) Curd (and sour milk) contains lactic acid. Citric is in lemons, acetic in vinegar, tartaric in tamarind — useful to keep these source-pairs straight.

4. (A) Vinegar is a dilute solution of acetic acid. (Formic acid is in ant stings; tartaric in tamarind; HCl in the stomach.)

5. (D) Tamarind is sour because of tartaric acid. (Lactic = curd, citric = citrus, acetic = vinegar — the distractors are the other common food acids.)

6. (B) Milk of magnesia (magnesium hydroxide) is a base, used as an antacid. Lemon juice, vinegar and curd are all acids.

7. (A) The stomach makes hydrochloric acid (HCl), which helps digest protein and kills germs in food. Sulphuric acid is industrial; citric and carbonic are weak food/drink acids.

8. (C) Litmus is extracted from lichens (a fungus–alga partnership) and is the simplest natural acid–base indicator. Phenolphthalein and methyl orange are synthetic; baking soda is a base, not an indicator.

9. (C) Blue litmus turning red is the test for an acid. (Bases turn red litmus blue; neutral solutions change neither.)

10. (C) No change in either litmus paper means the sample is neutral — neither acidic nor basic (e.g. distilled water, salt water).

11. (A) Turmeric is a natural indicator that turns red in a base. Soap solution is basic, so the turmeric stain turns red — the everyday "yellow stain turns red where soap touches it" observation.

12. (B) Phenolphthalein is colourless in acids/neutral and pink in a base. A pink result therefore means the solution is basic.

13. (A) The reaction of an acid with a base is called neutralisation; their opposite properties cancel.

14. (D) Acid + base → salt + water. (Hydrogen is released when acids react with metals, not in plain neutralisation.)

15. (C) An ant sting injects an acid (formic acid). Rubbing on a mild base — baking soda paste — neutralises it and relieves the pain. Lemon juice and vinegar are acids (they would make it worse).

16. (C) Acidic soil is corrected by adding a base. Quicklime/slaked lime (calcium oxide/hydroxide) neutralises the excess acid. Adding any acid would make it worse.

17. (B) Acidic effluent must be neutralised before release. Treating it with a base such as slaked lime brings it toward neutral and protects river life. Adding more acid, salt or sugar does not neutralise it.

18. (D) Acidity is excess stomach acid. An antacid is a mild base; it neutralises the extra acid and stops the burn. It is not an acid, sugar or salt.

19. (A) Common salt, $NaCl$, is the salt formed when hydrochloric acid (HCl) is neutralised by sodium hydroxide ($NaOH$): $HCl + NaOH \rightarrow NaCl$ + water.

20. (C) Distilled water is neutral, so it changes neither blue nor red litmus. (Litmus does not turn "green" — that is universal indicator, a higher-class idea.)

21. (D) China rose indicator turns green in a base (and dark pink/magenta in an acid). A light-green result therefore means the solution is basic.

22. (B) Phenolphthalein is colourless in acid and pink in base. Lemon juice (acid) → colourless; soap solution (base) → pink. Order: colourless, then pink.

23. (B) Blue litmus turns red only in acids. Of the three, only lemon juice is acidic; soap is basic and sugar solution is neutral. So exactly 1 turns blue litmus red.

24. (C) Sulphur dioxide (SO_2) from burning fuels dissolves in rain to form sulphuric acid — the main cause of acid rain. Oxygen, nitrogen and carbon monoxide are not the acid-formers here.

25. (A) Sodium hydroxide is a strong base, so it turns red litmus blue. (It does not turn blue litmus red — that is an acid's behaviour.)

26. (C) Phenolphthalein turns pink only in a base, and a base also turns red litmus blue. A liquid that turns phenolphthalein pink yet leaves both litmus papers unchanged is self-contradictory — the results don't agree, so the report is inconsistent.

27. (C) China rose is dark pink (magenta) in acid and green in base. Tube 1 (magenta) is an acid; tube 2 (green) is a base. Order: acid, then base.

28. (A) Vinegar is an acid and baking soda is a base; mixing them is a neutralisation that also releases gas (the fizz). So a base is reacting with an acid.

29. (A) Acid → base → neutral, in order, is lemon juice (acid), soap (base), distilled water (neutral). The other lists put the families in the wrong order.

30. (B) A stinging bee or ant injects formic acid. (HCl is the stomach acid; tartaric and carbonic are food/drink acids.)

31. (D) Milk of magnesia is a base — bases taste bitter and feel soapy. (Sour belongs to acids; the other pairs don't match a base.)

32. (B) Turning red litmus blue means the solution is a base. A base also turns turmeric paper red. It would not turn blue litmus red (that's acidic), and phenolphthalein turns pink (not colourless) in a base; turmeric does not turn green.

33. (D) Phenolphthalein is a synthetic, lab-made indicator. Litmus, turmeric and china rose are all natural indicators.

34. (B) Citrus fruits (orange, lemon, mosambi) are sour because of citric acid.

35. (C) When an acid and a base are mixed in exactly neutralising amounts, the result is neutral — it changes neither litmus paper.

36. (B) Turning red litmus blue is the test for a base, so ammonia is basic in nature (it forms ammonium hydroxide in water).

37. (A) HCl + NaOH → NaCl + water. Apart from water, the product is sodium chloride (common salt). The reactants HCl and NaOH are used up, not produced.

38. (A) Neutralisation is exothermic — heat is released and the mixture warms up. It does not freeze, turn into metal, or smell more strongly of acid (the acid is being consumed).

39. (B) Blue litmus turns red only in acids. Lemon juice and vinegar turn it red (acidic); soap and distilled water do not. So 2 of the four are acidic.

40. (B) Turmeric turns red only in a base; if it stays yellow, the sample is not basic. It could be acidic or neutral — turmeric alone cannot tell those two apart — so the safe conclusion is "not basic."

41. (B) The soil is too basic, so it needs something mildly acidic. Decaying organic matter (compost) is mildly acidic and lowers the pH gently. Slaked lime, NaOH and baking soda are all basic and would make it worse.

42. (D) Turmeric turns red in bases — that statement is correct. The others are wrong: phenolphthalein is colourless (not pink) in acids; blue litmus turns red in acids (not bases); china rose does change colour between acids and bases.

43. (D) Inside ENO, citric acid (acid) and sodium bicarbonate (base) react when wet: acid + base → salt + water + CO_2. The fizz is the CO_2 gas escaping — a neutralisation that releases a gas.

44. (D) Blue litmus turning red means acidic; phenolphthalein staying colourless is also consistent with acidic (it is only pink in bases). Both results agree on acidic, so the solution is an acid.

45. (C) Feeling soapy and turning red litmus blue are base-only properties. Sour taste and rough/sour behaviour belong to acids; turning blue litmus red is also an acid property.

46. (A) Sour taste and turning blue litmus red are both acid signatures, and they agree — so the liquid is an acid.

47. (D) The pink colour shows the soap solution is basic. Adding lemon juice (acid) neutralises the base; once enough acid is added the solution is no longer basic, so phenolphthalein loses its pink and turns colourless.

48. (D) Relieving an ant sting (acid) with baking soda (base) is a neutralisation reaction. A glowing torch, melting ice and a spinning fan involve no acid–base neutralisation.

49. (C) Sulphuric acid (H_2SO_4) is the most-used industrial acid — in fertilisers, car batteries and detergents. Citric, acetic and lactic are weak food acids, not industrial workhorses.

50. (A) A base turns red litmus blue — that is liquid X. Liquid Y (turns blue litmus red) is an acid; liquid Z (changes neither) is neutral. So X is the base.

51. (B) Turmeric turns red only in a base, so the sample is basic. A base turns red litmus blue. It would not turn blue litmus red, would not leave litmus unchanged, and bases taste bitter, not sour.

52. (D) Sugar dissolved in water is neutral, so it changes neither litmus paper and leaves phenolphthalein colourless. The "no change in either litmus" result is the correct one.

53. (C) Sodium hydroxide is a strong base, not an acid — so "sodium hydroxide — acid" is the wrong pairing. Baking soda (base), vinegar (acid) and lemon juice (acid) are all correctly paired.

54. (B) Marble and limestone are calcium carbonate, a base. The acid in acid rain reacts with and slowly eats away this base — that is why monuments like the Taj Mahal are damaged.

55. (C) A neutral solution changes neither litmus paper and leaves phenolphthalein colourless. The other option-sets mix in acid or base behaviour, so they don't describe a neutral solution.

56. (D) Chalk (calcium carbonate) is a mild base; added to dilute acid it partly neutralises the acid and releases carbon-dioxide gas, so you see fizzing. It cannot make the acid stronger.

57. (D) The question states methyl orange is yellow in bases and red in acids. In a basic solution it shows yellow.

58. (D) Many neutral solutions besides water — salt water, sugar water — also leave litmus unchanged. So "must be water" is wrong; leaving litmus unchanged only proves the liquid is neutral, not that it is pure water.

59. (A) China rose is magenta (dark pink) in acid and green in base. The first solution (magenta) is an acid; the second (green) is a base. Order: acid then base.

60. (A) Baking soda is a base. It neutralises the excess (acidic) stomach acid, forming salt and water (with some gas), which relieves the burning. It does not add acid, increase acidity, or merely coat the stomach.

Solutions complete: 60 / 60.